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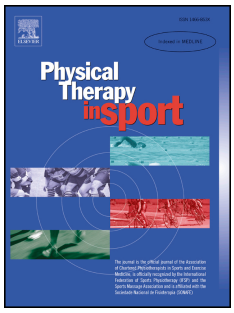

Hip adduction and abduction strength values in NCAA Division I American football players: strength values and a comparison across position groups.

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PII: S1466-853X(25)00030-6

DOI: https://doi.org/10.1016/j.ptsp.2025.01.004

Reference: YPTSP 1776

To appear in: *Physical Therapy in Sport*

Received Date: 11 October 2024

Revised Date: 22 January 2025

Accepted Date: 23 January 2025

Please cite this article as: O' Connor, C., Renaghan, E., Bennett, A., Ferrer, L., Damodaran, R., Geraghty, P., Feigenbaum, L.A., Hip adduction and abduction strength values in NCAA Division I American football players: strength values and a comparison across position groups., *Physical Therapy in Sports*, https://doi.org/10.1016/j.ptsp.2025.01.004.

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Title

Hip adduction and abduction strength values in NCAA Division I American football players: strength values and a comparison across position groups.

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Title

Hip adduction and abduction strength values in NCAA Division I American football players: strength values and a comparison across position groups.

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Abstract

Objectives: To report hip adduction, abduction, and adduction:abduction strength ratio values in NCAA Division I American football players, and to compare strength values across position groups.

Design: Cross-sectional study.

Setting: University training facility.

Participants: 85 male football players.

Main Outcome Measures: Isometric hip adduction and abduction strength values in the 0° hip-flexion long-lever and 0° hip-flexion short-lever testing positions, using a ForceFrame.

Results: Hip strength values are presented across two testing positions. ANOVA revealed significant differences ($p \leq 0.05$) in hip strength between position groups, with medium-to-large effect size (n^2 = 0.116–0.284). Skill-position and mid-position players demonstrated significantly greater hip adduction and abduction relative strength in both testing positions, when compared to lineman-position players. In addition, skill-position and mid-position players demonstrated significantly greater hip adduction:abduction strength ratios (0.19-0.20, 24-25%) in the long-lever testing position, when compared to lineman-position players, with large effect size (n^2 = 0.178). No significant differences in strength were observed between skill-position and mid-position players. Hip adduction:abduction strength ratios ranged from 0.68–0.88 in the long-lever testing position and 1.05–1.09 in the short-lever testing position.

Conclusion: Significant differences in relative strength exist between position groups of NCAA Division-I football players, in hip adduction, abduction and adduction:abduction ratio.

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Key Words: Groin Strength, Hip Strength, Hip Adductors, Hip Adduction:Abduction Ratio.

Abstract Word Count: 198

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INTRODUCTION

Hip and groin injuries are commonly sustained by field-sport athletes and have been reported to account for as high as 23% of all injuries in some field-sport cohorts.[1-7] Much of the research to date pertains to sports that involve a lot of kicking, such as soccer,[2, 5, 6] Australian football,[1] Gaelic football,[4] and rugby union,[7] while hip and groin injury data relating to American football cohorts is generally less available.[8, 9] The existing literature reports the overall rate of hip injuries amongst NCAA Division I football players as 5.18 per 10,000 athlete exposures. It has been reported that hip/groin injuries account for 6% of all injuries sustained in NCAA Division I football, and 7.5% of all injuries sustained in practice.[8] Specifically hip adductor, hip abductor and hip flexor muscles strains account for 75% of all hip injuries in collegiate football.[9] However, these statistics likely do not represent the full extent of the issue, as it has been reported that many field-sport athletes continue to participate in their sport despite the presence of persistent groin pain.[10] Further to this, athletes who return from a time-loss groin injury, show reduced physical function upon their return to sport.[11] Additionally, with reported re-injury rates as high as 29% in groin-injured athletes,[12] it is clear that hip and groin injuries pose a significant problem to athletic trainers and sports medicine practitioners within collegiate American football populations.

Recent systematic reviews have reported that lower isometric hip adductor strength is one of the primary intrinsic risk factors for hip/groin injuries.[13, 14] Athletes with lower hip adduction strength are up to four times more likely to suffer from hip and groin pain.[2] Moreover, several studies have reported a strong association between reduced pre-season hip adduction strength and a higher incidence of time-loss groin injury in field-sport athletes. Reduced hip abductor strength,[15] and reduced hip adductor:abductor strength ratio[3] have also been suggested as risk factors for hip/groin injury in field sport athletes. Consequently, the periodic assessment of hip adduction and abduction strength values have become common practice in many field sport populations. However, there remains little if any published data on the hip strength qualities of collegiate American football players, nor does the assessment of hip adduction and abduction strength appear common practice within the sport. Furthermore, the absence of reported normative data for hip adduction squeeze and abduction press

strength in collegiate football players makes it difficult to assess the relationship between these hip strength metrics and the risk of sustaining a hip/groin injury. Therefore, risk assessments and injury profiling is difficult within this population. Considering that the average time lost from sport by a field-sport athlete sustaining a groin injury has been reported as high as 27 days,[7] teams are potentially losing a significant amount of player availability to hip/groin injuries, which is in part due to an inability to accurately assess an athletes risk of hip/groin injury. Accurate screening for athletes at high risk of hip/groin injury could reduce injury incidence and increase player availability throughout the season, as has been implemented in other field-sport cohorts.[16] Moreover, the use of replicable testing protocols, with reliable and valid testing devices, would enable hip strength assessments aimed at categorising an athlete's hip adduction and abduction strength relative to their population and other field-sport populations.

Much of the current literature documenting hip strength values use inconsistent testing devices, specifically the hand-held dynamometer[1, 5, 17] and the sphygmomanometer[4, 18, 19] – both of which are susceptible to error, given the lack of external fixation amongst other issues. The hand-held dynamometer has been shown to be influenced by the strength and sex of the tester,[20] whereas the sphygmomanometer has been shown to have a ceiling effect,[19, 21] with its validity as a measure of hip strength recently called into question. More recently, the ForceFrame Strength Testing System, (Vald Performance) has demonstrated 'excellent' test re-test reliability.[21, 22] It has been suggested as the 'gold standard' of isometric hip strength assessment as it provides external fixation in the form of a fixed frame, has the ability to provide hip adduction and abduction strength values for each limb independently while using a bilateral assessment, and does not rely on tester strength or experience.[21] In addition, a standardised testing protocol has been used in recent studies collecting reference hip strength data from field-sport athletes across soccer,[23] rugby union[24] and Gaelic football.[24]

Therefore, the primary aim of this study was to report hip adduction and abduction strength values in National Collegiate Athletic Association (NCAA) Division I football players in the 0° hip flexion long-lever testing position and the 0° hip flexion short-lever testing position, using the ForceFrame Strength

Testing System. The secondary aim was to compare the strength values of hip adduction and abduction between position groups, to examine if differences in hip strength exist between position groups.

METHODS

Participants

Eighty-five male athletes (mean ± SD: age = 20.0 ± 1.79 years; range 18-26 years; body mass = 108.3 ± 23.69 kg; height = 188.7 ± 6.57 cm) volunteered to partake in this study. All participants were NCAA Division I football players, and were playing for the same team competing in the Atlantic Coast Conference (ACC). Ethical approval was obtained through the Human Ethics Committee (20180810), and information about the study was provided to all participants prior to providing informed consent. Players with a previous hip/groin injury were included in the study to provide an accurate representation of the playing population at this level. All participants had previously been exposed to team-based strength and conditioning programmes and were training regularly with the team at the time of data collection. Data collection was undertaken at the University training facilities, before a morning practice, at the start of the summer pre-season training period, three months prior to the start of the competitive season. Physical assessments, inclusive of strength assessments, were regularly conducted as part of the teams normal practice. Participants were not given any specific hydration-related or nutrition-related instructions prior to testing.

Table 1 here.

Testing Protocol

Upon arrival at the training facility, subjects were informed of the testing procedure and given a demonstration of the strength tests, before being asked a series of short questions. The questions consisted of information such as age, playing position and dominant leg (as defined by the preferred kicking foot when kicking a ball).[21] After completing the questionnaire, anthropometric data in the form of weight, height and limb length measurements were obtained. Leg length was obtained from

both legs independently and measured using a tape measure from the anterior superior iliac spine (ASIS) to the medial malleolus, whilst the participant lay in a supine position.[25]

The strength testing protocol comprised of maximum isometric hip adduction squeeze and abduction press strength testing in the 0° hip flexion long-lever testing position, and 0° hip flexion short-lever testing position, using the ForceFrame Strength Testing System (VALD Performance, Brisbane, Australia). The ForceFrame has established excellent reliability and validity in both long-lever and short-lever testing positions, as a measurement of hip adduction squeeze and abduction press isometric strength in field-sport cohorts.[21, 22] The order of testing position were randomized, whilst the order of assessments (i.e., hip adduction squeeze and then hip abduction press, or vice versa) were alternated with each repetition. The subjects completed both adduction squeeze and abduction press tests in one testing position, before progressing to the next testing position. Subjects were asked to perform a maximum voluntary isometric contraction for 5 seconds, by bilaterally pushing their femoral medial (adduction) and lateral (abduction) condyles for short-lever testing, or medial (adduction) and lateral (abduction) malleoli for long-lever testing, against the force pads.[21] The test procedure required two maximum-effort trials for both adduction squeeze and abduction press, in each testing position, in order to accommodate time-efficient testing of the subjects. The peak value obtained across the two trials, for both adduction squeeze and abduction press testing, was used in the data analysis. Subjects were allowed a 10-second rest period between repetitions and a 60-second rest period between each testing position. Throughout all testing, the tester provided strong verbal encouragement to elicit maximum effort from the athletes. Subjects were not informed of their scores during testing, nor were they able to see their scores throughout the duration of the testing protocol. Pain to the hip/groin area was recorded during both adduction squeeze testing and abduction press testing in both testing positions, using a 0-10 numeric pain rating scale (NPRS) with 0 representing no pain and 10 representing maximal pain. Subjects were asked after each trial to categorise their pain using a visual NPRS chart.

Force data from the ForceFrame were collected via custom made software (Scoreboard, VALD Performance, Brisbane, Australia) and transferred manually to a personal computer. The peak force for

both limbs (dominant and non-dominant - as defined by the preferred kicking foot when kicking a ball) for each test and testing position was determined automatically through the Scoreboard software and expressed as absolute (N) and relative (N/kg) strength values. Normalised strength values (Nm/kg) with respect to limb length were calculated as Newton meters per kilogram of body mass, for each limb independently, for the long-lever testing position. Bilateral values (N, N/kg and Nm/kg) were calculated and represent a mean of dominant limb peak force and non-dominant limb peak force. Limb asymmetry represents the mean of the percentage difference in peak force between dominant and non-dominant limbs, per each participant. Strength data was presented as mean ± SD.

Statistical Analysis

Where appropriate, a Shapiro-Wilk test was used to test the normal distribution of the data. Initially, participant data was separated into two groups: players with pain during strength testing vs. players with no pain during strength testing. Independent samples *t*-tests were used to compare the mean values of hip adduction squeeze and abduction press strength between groups. Results revealed no significant differences ($p = 0.332$) between groups, so players with pain during testing were included in the dataset for further analyses. Participant data was then separated into three positional groups, with 'lineman position' representing all positions of defensive line (defensive tackle and defensive end) and offensive line (center, tackle and guard), 'mid position' representing the positions of tight-end, running back and linebacker, and 'skill position' representing the positions of quarterback, wide-receiver, defensive back, punter, kicker, and snapper. A one-way analysis of variances (ANOVA) was used to compare mean values age, height and body mass between the three position groupings. An ANOVA was then used to compare mean values of absolute strength (N), relative strength (N/kg) and normalised strength (Nm/kg) for hip adduction squeeze, hip abduction press and hip adduction:abduction ratio between the three position groupings. Comparisons of normalised strength (Nm/kg) were only possible in the 0° hip flexion long-lever testing position, while comparisons of absolute strength and relative strength values were performed for both testing positions. Statistical significance was accepted at an alpha level of $p < 0.05$. For effect size, the partial eta squared (n^2) was reported, with values of 0.01, 0.06, and 0.14 considered indicative of small, medium, and large effect sizes respectively, in accordance with the most

recent guidelines.[26] A post-hoc analysis with Bonferroni correction was used to examine specific differences in strength between position groupings. Statistical significance was accepted at an alpha level of $p < 0.05$. All analyses were performed using IBM SPSS Statistics Version 26 (IBM Corp., Armonk, NY, USA).

RESULTS

Subject characteristics are reported in Table 1. ANOVA revealed no significant differences in age (F = 0.369, $p = 0.692$, $n^2 = 0.009$) between position groups, however there were significant differences in both height (F = 12.577, $p = <0.001$, $n^2 = 0.235$) and body mass (F = 112.145, $p = <0.001$, $n^2 = 0.732$) between position groups. The lineman position group demonstrated significantly greater height compared to the mid position group ($p = 0.046$) and the skill position group ($p = <0.001$). Similarly, the lineman position group demonstrated significantly greater body mass compared to the mid position group ($p = <0.001$) and the skill position group ($p = <0.001$), while the mid position group also demonstrated significantly greater body mass than the skill position group ($p = 0.001$).

Strength values for maximum isometric hip adduction squeeze, hip abduction press, and hip adduction:abduction strength ratio are presented in Table 2, for absolute (N), relative (N/kg) and normalised strength (Nm/kg). NCAA Division I football players demonstrated lower hip adduction squeeze strength relative to hip abduction press strength in the 0° hip flexion long-lever testing position, and higher hip adduction squeeze strength relative to hip abduction press strength in the 0° hip flexion short-lever testing position.

Table 2 here.

A comparison of absolute strength values between position groups is reported in Table 3. ANOVA revealed significant differences in hip abduction press strength and hip adduction:abduction strength ratio in the 0° hip flexion long-lever testing position, between position groups, with medium ($n^2 = 0.126$) and large ($n^2 = 0.178$) effect size respectively. No other differences in absolute strength for hip adduction

squeeze, hip abduction press or hip adduction:abduction strength ratio, between position groups, reached statistical significance.

Table 3 here.

Meanwhile, a comparison of relative strength values and normalised strength values between position groups is reported in Table 4. ANOVA revealed significant differences in hip adduction squeeze and abduction press relative strength between positions groups, in both testing positions, with medium-to-large effect size (n^2 ranging from 0.116 – 0.284). The skill position group demonstrated significantly greater hip adduction squeeze relative strength (0.65 – 0.96 N/kg, 31 – 45%) and hip abduction press relative strength (0.37 – 1.01 N/kg, 20 – 33%) across both testing positions, when compared to the lineman position group. Similarly, the mid position group demonstrated significantly greater hip adduction squeeze relative strength (0.59 – 0.88 N/kg, 28 – 41%) across both testing positions, and significantly greater hip abduction press relative strength (0.83 N/kg, 28%) in the 0° hip flexion short-lever position, when compared to the lineman position group. In addition, the mid position group demonstrated greater hip abduction press relative strength (0.28 N/kg, 15%) than the lineman position group in the 0° hip flexion long-lever position, however this difference in strength did not reach significance ($p = 0.081$). There were no statistically significant differences in strength between the mid position group and the skill position in any test or testing position, however the skill position group demonstrated numerically higher hip adduction squeeze and hip abduction press strength values across all tests when compared to the mid position group. Finally, the same positional differences in strength were evident when analysing the normalised strength values in the 0° hip flexion long-lever position, as were evident when analysing the relative strength values in the same testing position, suggesting lever length did not impact the comparative analysis (Table 4). Normalised strength values for hip adduction squeeze, hip abduction press and hip adduction:abduction strength ratio are presented in Figure 1, demonstrating the normal distribution of data.

Table 4 here.

Figure 1 here.

DISCUSSION

The primary aim of this study was to report hip adduction and abduction strength values in National Collegiate Athletic Association (NCAA) Division I football players in the 0° hip flexion long-lever testing position and the 0° hip flexion short-lever testing position, using the ForceFrame Strength Testing System. Whilst reference values exist for hip adduction squeeze and abduction press for soccer, Gaelic football, rugby union, and Australian football players, this study is the first to report such strength values in NCAA Division I American football players.

Hip strength values in two testing positions, inclusive of unilateral and bilateral values should aid practitioners in profiling American football players hip strength. In addition, the provision of strength data relative to body mass, lever length and position grouping provides practitioners with further insights that may aid in identifying athletes with hip strength deficits relative to their respective cohort. With an absence of comparative data available within American football cohorts, the strength data we report in this population can be compared to recently reported data in field-sports with similar movement demands to American football, specifically Gaelic football, rugby union and soccer.[24, 27] The use of a consistent testing protocol, testing positions and testing equipment allows for an accurate and valid comparison of hip strength data between cohorts, which provides insights to the hip strength profile of the American football players in the current study.[24] Notably, the hip adduction squeeze strength, abduction press strength and adduction:abduction strength ratio reported in American football players, irrespective of position grouping, are all markedly lower than those of male Gaelic football players, rugby union players and soccer players.[24, 27] Meanwhile, the hip adduction:abduction strength ratios observed in the present study are also markedly lower than comparable field-sport cohorts.[24] In particular, the hip adduction:abduction strength ratio observed in the 0° hip flexion long-lever position (0.81 ± 0.22) is notably low. Research across field-based sports indicates that healthy athletes typically demonstrate hip adduction:abduction strength ratios closer to, or greater than, 1.00 when tested in the 0° hip flexion long-lever testing position.[3, 18, 22, 24] It has been reported that an athlete is 17 times more

likely to sustain a hip adductor muscle injury if their pre-season hip adduction:abduction ratio is lower than 0.80.[3] Indeed, a hip adduction:abduction strength ratio lower than 0.90 has been used as an 'alert' to identify athletes with groin problems and athletes at higher risk of a groin injury, when monitoring hip strength during a playing season.[16, 28] Therefore, the group-level hip adduction:abduction strength ratio of 0.81 ± 0.22 observed the present study, and the hip adduction:abduction strength ratio of 0.68 ± 0.21 observed in the lineman position group, suggests the hip strength ratio of American football players is significantly lower than many other field sports. How this pertains to hip/groin injury risk, however, has yet to be established.

A potential explanation for the notably low hip strength values, relative to data from similar field-sports, reported in the current study is the different strength and conditioning practices that may exist between these sports. With periodic screening for hip adduction squeeze and abduction press strength common practice throughout Gaelic football, rugby union and most notably soccer, it is possible that greater emphasis is put on training these strength qualities within these sports. Nevertheless, the relatively lower hip adduction squeeze and abduction press strength observed in American football players may have implications towards hip/groin injury risk. Considering the abundance of research associating lower hip adduction squeeze strength and lower hip abduction press strength in fields-port athletes to a higher risk of hip/groin injury,[1, 6, 15] it is likely that American football players would benefit from increased strength in the hip adductor and abductor muscles

Meanwhile, the secondary aim of this study was to compare the strength values for hip adduction and abduction between position groups, and to examine if differences in hip strength exist between position groups. Whilst we observed very comparable absolute hip strength values across position groups, we found significant differences in hip adduction squeeze and hip abduction press strength between position groups, when strength was expressed relative to body mass or lever length. Given the differences in movement patterns and demands between position groups in American football,[29] it is not surprising that statistically significant differences in relative strength (N/kg) and normalised strength (Nm/kg) were revealed between position groupings. Skill position players (specifically the defensive

back position and the wide receiver position) and mid position players (inclusive of the tight end position, running back position, and linebacker position) all typically perform significantly greater total running distance, high-velocity distance, accelerations, decelerations – and consequently a greater volume of high-speed change of directions – when compared to lineman position players.[29, 30] These movements have been shown to involve high muscle activity in the hip adductors and hip abductors.[31-33] Specifically, the hip adductor muscles stabilise the hip during the stance phase of running, and decelerate the body and pelvis during change of direction running,[34] with high hip adductor forces evident during these tasks. In addition, high muscle activity in the hip abductor muscles has also been demonstrated during change of direction running.[33] Therefore, it is possible that the greater hip adduction squeeze and abduction press relative strength demonstrated by players in the skill position and mid position groups is reflective of the running, sprinting and change of direction demands associated with these positions.

The significantly lower hip adduction squeeze and abduction press strength observed in lineman position players is still however surprising, considering players in these positions typically encounter a greater volume of collisions[35] where the hip adductor muscles, amongst others, contract isometrically to maintain posture and position against the opponents opposing force. Additionally, and perhaps most notably, when strength was expressed as an absolute value (N) and not relative to body mass or lever length, lineman position players still demonstrated lower hip adduction squeeze strength (147.1 ± 45.98 N) in the 0° hip flexion long-lever testing position compared to mid position players (171.8 ± 42.59 N) and skill position players (157.7 ± 44.22 N) respectively. Therefore, the significantly lower hip adduction squeeze strength observed in lineman position players compared to the skill position and mid position groups, cannot simply be attributed to differences in body mass. Considering this, and the well-established relationship between lower hip adduction strength and an increased groin injury risk, it is likely that the lineman position group in particular could benefit from hip adductor specific strengthening exercises to increase hip adduction squeeze strength in long-lever positions.

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Finally, this study has many limitations. Firstly, the strength data reported in this study was obtained at the start of the summer training camp period, and considering that previous research has reported significant changes in the hip adduction squeeze and abduction press strength values of field-sport athletes across various seasonal timepoints,[16, 18] it is possible that the strength values reported in the current study may change as the season progresses. However, as this is the first study to report hip adduction squeeze and abduction press strength values in American football players, it is not yet established if changes in hip strength occur throughout a season. Another potential limitation of this study is that the data we report reflects a convenience sample from a single NCAA Division I football programme, and may indeed be reflective of the hip strength training practices in place within that programme. Whilst our study should be seen as preliminary and insightful, future research seeking to replicate this study should aim to include a much larger sample size inclusive of several Division I programmes.

CONCLUSION

Hip strength values for adduction squeeze, abduction press and adduction:abduction ratio are provided for NCAA Division I American football players, across two testing positions. A comparison across position groupings revealed medium-to-large differences in hip adduction squeeze and abduction press relative strength position groupings, in NCAA Division I football players. Skill position players and mid position players demonstrate significantly greater hip adduction squeeze and abduction press relative strength when compared to lineman position players. In addition, skill position players and mid position players demonstrate significantly greater hip adduction:abduction strength ratio in the 0° hip flexion long-lever testing position, when compared to lineman position players. No significant differences in hip strength were evident between skill position players and mid position players. At group-level, American football players demonstrate lower hip adduction squeeze and abduction press relative strength, and notably lower hip adduction:abduction strength ratios, when compared to other field-sport cohorts.

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Highlights

- Hip strength values were reported for NCAA Division I football players.
- Significant differences in relative hip strength exist between position groups.
- Lineman position players demonstrate notably weaker hip strength to mid and skill position players.

Declaration of Interest

None

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Figure Legend

Figure 1: Maximum isometric hip adduction squeeze strength, abduction press strength, and adduction:abduction strength ratio (Nm/kg) in NCAA Division I football players, in the 0° hip flexion long-lever testing position.

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Table 1. Characteristics of study participants.

Descriptive	All players Mean ± SD	Lineman position [a] Mean ± SD	Mid position [b] Mean ± SD	Skill position [c] Mean ± SD
Playing level	NCAA Division I	NCAA Division I	NCAA Division I	NCAA Division I
Number of participants	85	31	21	33
Training frequency	5-6 sessions per week	5-6 sessions per week	5-6 sessions per week	5-6 sessions per week
Age (years)	20.0 ± 1.79	20.2 ± 1.86	19.8 ± 1.87	20.0 ± 1.72
Weight (kg)	108.3 ± 23.69	134.1 ± 16.92	101.4 ± 11.14[ǂ]	88.5 ± 6.96[ǂ #]
Height (cm)	188.7 ± 6.57	192.5 ± 5.78	188.4 ± 6.94[ǂ]	185.2 ± 5.02[ǂ]
Right foot DOM (n)	76 (89%)	27 (87%)	19 (90%)	30 (91%)
Left foot DOM (n)	9 (11%)	4 (13%)	2 (10%)	3 (9%)
Hip/groin pain during adduction testing (n)	6 (7%)	4 (13%)	1 (5%)	1 (3%)
Hip/groin pain during abduction testing (n)	5 (6%)	2 (6.5%)	2 (10%)	1 (3%)

SD: Standard Deviation; DOM: Dominant.

ǂ significantly different ($p = \leq 0.001$) from lineman position group.

significantly different ($p = \leq 0.001$) from mid position group.

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Table 2: Reference values for maximum isometric hip adduction squeeze strength, abduction press strength and adduction:abduction strength ratio in NCAA Division I football players (n = 85).

Leg	0° Long-lever testing position			0° Short-lever testing position		
	Adduction Mean ± SD	Abduction Mean ± SD	Ratio Mean ± SD	Adduction Mean ± SD	Abduction Mean ± SD	Ratio Mean ± SD
Absolute Strength (N)						
Dominant	157.3 ± 45.23	203.5 ± 50.90	0.80 ± 0.24	337.5 ± 124.59	326.7 ± 98.00	1.06 ± 0.34
Non-Dominant	157.5 ± 45.68	199.8 ± 49.48	0.81 ± 0.23	338.0 ± 124.93	322.0 ± 101.39	1.08 ± 0.35
Bilateral	157.4 ± 44.95	201.7 ± 49.46	0.81 ± 0.22	337.7 ± 123.98	324.4 ± 98.52	1.07 ± 0.34
Asymmetry	6.80 ± 5.90	6.68 ± 5.39	11.38 ± 10.66	6.24 ± 5.31	6.45 ± 5.47	9.94 ± 7.56
Relative Strength (N/kg)						
Dominant	1.54 ± 0.56	1.93 ± 0.47	0.80 ± 0.24	3.25 ± 1.30	3.15 ± 1.07	1.06 ± 0.34
Non-Dominant	1.54 ± 0.57	1.90 ± 0.48	0.81 ± 0.23	3.26 ± 1.33	3.10 ± 1.07	1.08 ± 0.35
Bilateral	1.54 ± 0.56	1.91 ± 0.47	0.81 ± 0.22	3.26 ± 1.31	3.12 ± 1.06	1.07 ± 0.34
Normalised Strength (Nm/kg)						
Dominant	1.56 ± 0.56	1.96 ± 0.46	0.80 ± 0.24	-	-	-
Non-Dominant	1.56 ± 0.56	1.93 ± 0.47	0.81 ± 0.23	-	-	-
Bilateral	1.56 ± 0.56	1.94 ± 0.46	0.81 ± 0.22	-	-	-

SD: Standard Deviation; Bilateral: Average between Dominant and Non-Dominant limb values; Asymmetry: % Difference between Dominant and Non-Dominant limb values; N: Newtons; N/kg: Newtons per kilogram of body mass; Nm/kg: Newtons meters per kilogram of body mass.

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Table 3: A comparison of bilateral maximum isometric hip adduction squeeze and abduction press absolute strength values (N) between position groupings in NCAA Division I football players (n = 85).

Testing position	Lineman position (n = 31)	Mid position (n = 21)	Skill position (n = 33)	ANOVA (one-way repeated measures)
Absolute Strength (N)	Mean ± SD	Mean ± SD	Mean ± SD	
Adduction (0° Long-Lever Position)	147.1 ± 45.98	171.8 ± 42.59	157.7 ± 44.22	F = 1.913 *p* = 0.154 n^2 = 0.045
Abduction (0° Long-Lever Position)	223.6 ± 53.96	199.3 ± 39.22	183.2 ± 44.07ǂ	F = 5.865 *p* = 0.004* n^2 = 0.126
Adduction:Abduction Ratio (0° Long-Lever Position)	0.68 ± 0.21	0.87 ± 0.19ǂ	0.88 ± 0.21ǂ	F =8.799 *p* = < 0.001* n^2 = 0.178
Adduction (0° Short-Lever Position)	348.5 ± 130.54	351.9 ± 125.72	319.3 ± 118.38	F = 0.603 *p* = 0.550 n^2 = 0.015
Abduction (0° Short-Lever Position)	328.9 ± 101.63	336.5 ± 101.17	312.6 ± 95.81	F = 0.420 *p* = 0.658 n^2 = 0.010
Adduction:Abduction Ratio (0° Short-Lever Position)	1.09 ± 0.34	1.09 ± 0.38	1.05 ± 0.32	F = 0.130 *p* = 0.879 n^2 = 0.003

SD: Standard Deviation; N: Newtons; Nm/kg: ANOVA: Analysis of variance; F: ANOVA; n^2: partial eta squared.

* denotes statistical significance ($p = \leq 0.001$).

ǂ significantly different ($p = \leq 0.001$) from lineman position group.

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Table 4: A comparison of bilateral maximum isometric hip adduction squeeze and abduction press relative strength values (N/kg) and normalised strength values (Nm/kg) between position groupings in NCAA Division I football players (n = 85).

Testing position	Lineman position (n = 31)	Mid position (n = 21)	Skill position (n = 33)	ANOVA (one-way repeated measures)
Normalised Strength (Nm/kg)	Mean ± SD	Mean ± SD	Mean ± SD	
Adduction (0° Long-Lever Position)	1.18 ± 0.45	1.75 ± 0.53‡	1.78 ± 0.49‡	F = 14.216 *p* = < 0.001* n^2 = 0.260
Abduction (0° Long-Lever Position)	1.77 ± 0.47	2.01 ± 0.42	2.06 ± 0.44‡	F = 3.615 *p* = 0.031* n^2 = 0.082
Relative Strength (N/kg)				
Adduction (0° Long-Lever Position)	1.13 ± 0.43	1.72 ± 0.53‡	1.78 ± 0.49‡	F = 16.843 *p* = < 0.001* n^2 = 0.294
Abduction (0° Long-Lever Position)	1.70 ± 0.45	1.98 ± 0.41	2.07 ± 0.45‡	F = 5.837 *p* = 0.004* n^2 = 0.126
Adduction (0° Short-Lever Position)	2.65 ± 1.03	3.53 ± 1.44‡	3.61 ± 1.29‡	F = 5.252 *p* = 0.007* n^2 = 0.116
Abduction (0° Short-Lever Position)	2.51 ± 0.85	3.34 ± 1.07‡	3.52 ± 0.99‡	F = 8.638 *p* = < 0.001* n^2 = 0.188

SD: Standard Deviation; N/kg: Newtons per kilogram of body mass; Nm/kg: Newton meters per kilogram of body mass; ANOVA: Analysis of variance; F: ANOVA; n^2: partial eta squared.

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* denotes statistical significance ($p = \leq 0.001$).

ǂ significantly different ($p = \leq 0.001$) from lineman position group.

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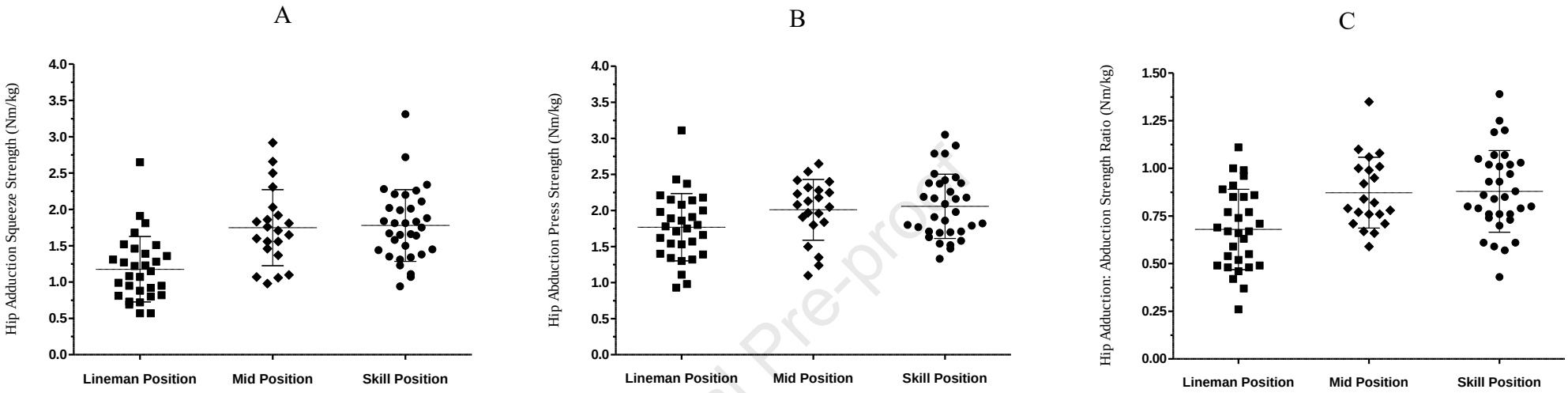
Figure 1: Maximum isometric hip adduction squeeze strength, abduction press strength, and adduction:abduction strength ratio values (Nm/kg) in NCAA Division I football players, in the 0° hip flexion long-lever testing position.



■ Lineman Position Players

◆ Mid Position Players

● Skill Position Players

A: Hip adduction squeeze strength

B: Hip abduction press strength

C: Hip adduction:abduction strength ratio

Mean ± SD represented with mean bar and accompanying error bars.

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Ethical Approval

Ethical approval was obtained through the Human Ethics Committee (20180810), and information about the study was provided to all participants prior to providing informed consent.

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Funding

None received.

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Declaration of Interest

None.

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